Chapter 2 Geological time

EXTRA WORKSHEET 2 ANSWERS: Dinosaurs and extinction

1 **a** K–T extinction, Cretaceous–Tertiary extinction

b About 65.5 million years ago**.**

2 Climate change, asteroid impact, volcanic eruption, change in food plants, sex ratio changes.

3 Extinction means the death of all organisms of a particular species, or mass extinction, meaning many species became extinct at the same time.

4 For: high levels or iridium an element rarely found in the Earth's crust, impact crater of about right age has been found.

Against: one study found that the Chicxulub asteroid hit the Earth about 300 000 years before the extinction. That research also suggested that the high iridium levels may be explained by volcanic activity.

5 Rock from the Deccan Traps has been dated to 65.5 million years ago. The eruptions would have released dust and other material, including iridium, into the air.

There were clearly changes in the Earth's climate at least half a million years before the extinction of dinosaurs, which corresponds with the date of eruption of the Deccan volcanoes.

6 Change in the type of food plants that grew; change the sex ratio of the dinosaur population which disrupted mating.

7 The dust might have blocked sunlight and so reduced photosynthesis in plants, reducing food supply of dinosaurs.

Copyright © Pearson Australia 2012 (a division of Pearson Australia Group Pty Ltd)
ISBN 978 1 4425 2366 1